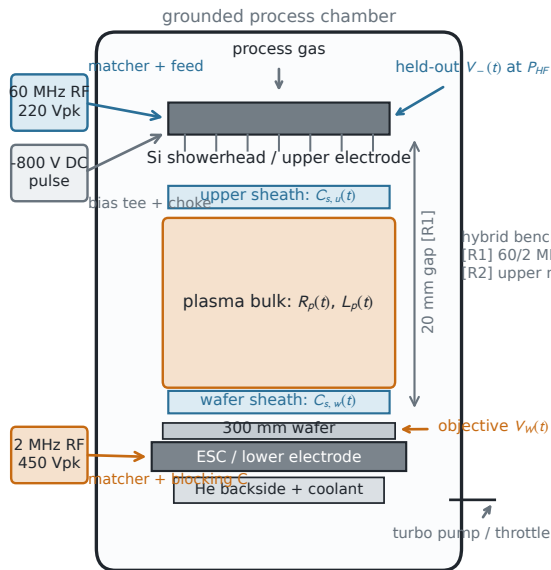


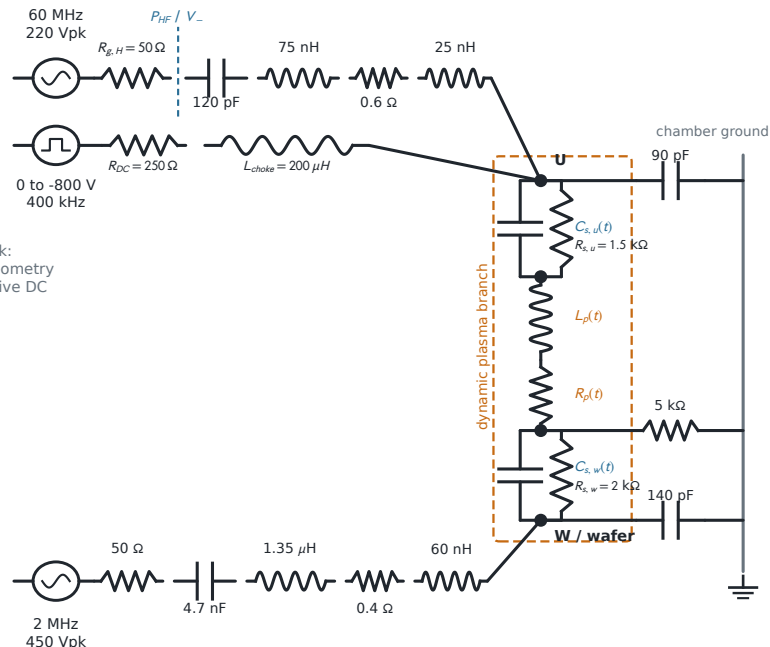
Literature-grounded etch-CCP topology and the solved electrical network

The chamber and schematic share U-to-W vertical order; this is a hybrid benchmark, not a drawing copied from one reactor.

(a) Chamber section and measurement planes **(b)** Solved ngspice electrical schematic



hybrid benchmark:
[R1] 60/2 MHz geometry
[R2] upper negative DC



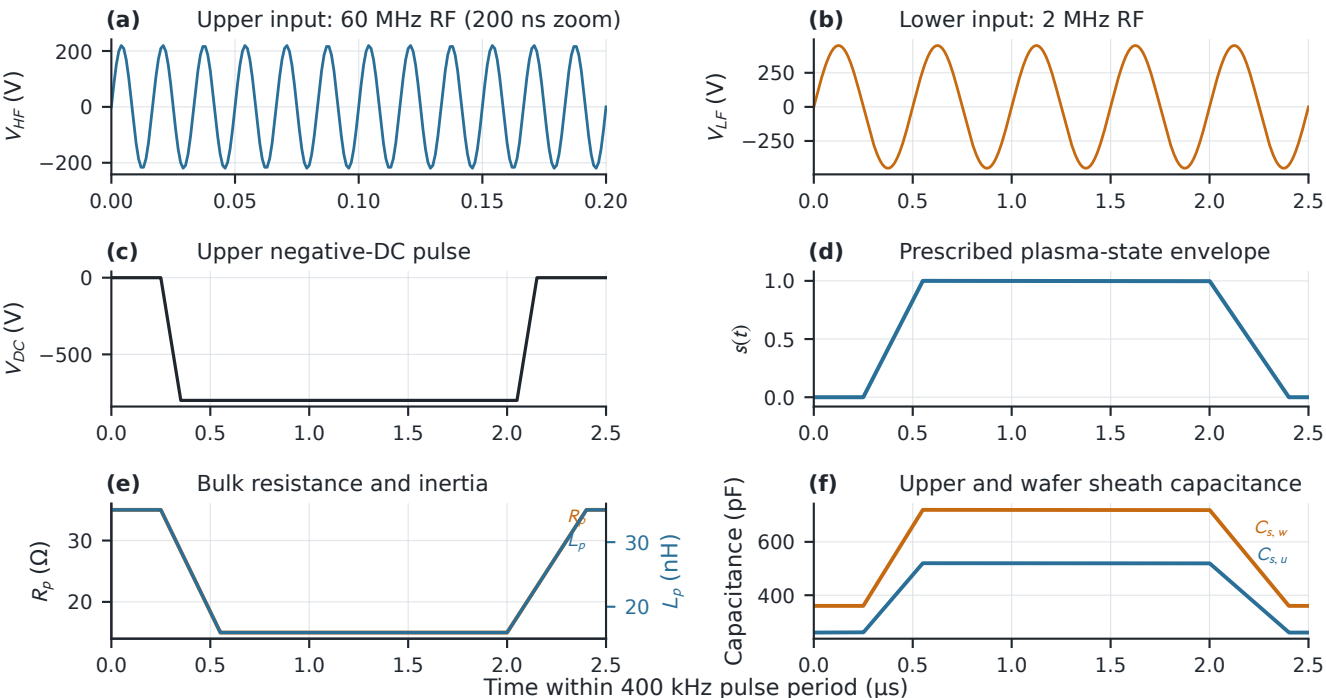
[R1] Kim et al., JJAP 54, 01AE07 (300 mm, 20 mm, upper 60 MHz/lower 2 MHz).

[R2] Yamaguchi et al., J. Phys. D 45, 025203 (negative DC on upper VHF electrode).

See p. 6 for source details, benchmark-specific assumptions, and the model boundary.

Known excitations and prescribed dynamic plasma elements

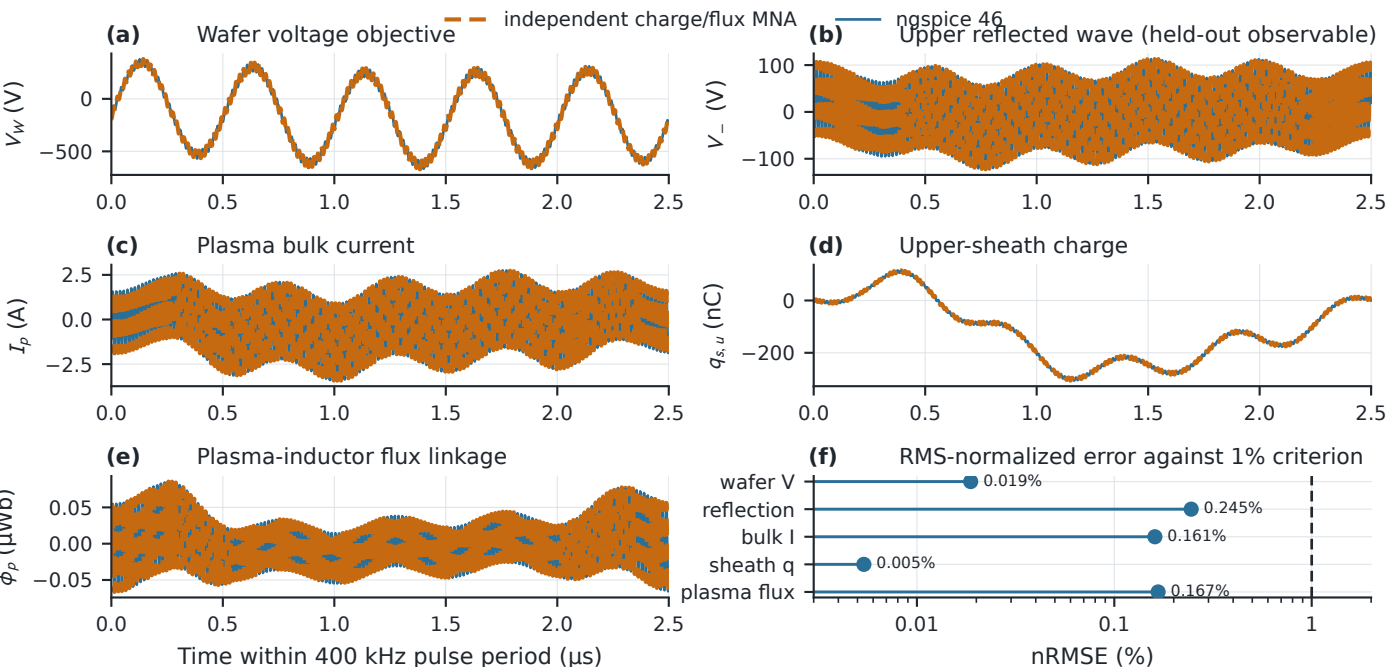
The prescribed envelope moves R_p : 35→15 Ω , L_p : 35→16 nH, $C_{s,u}$: 260→520 pF and $C_{s,w}$: 360→720 pF.



Constitutive laws used by both solvers: $q_s = C_s(t)v_s$ and $\phi_p = L_p(t)i_p$; therefore $i_s = dq_s/dt$ and $v_L = d\phi_p/dt$.

Forward validity: independent state equations and ngspice agree

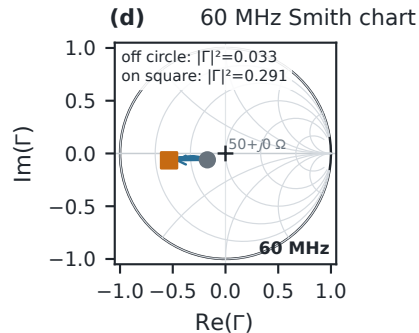
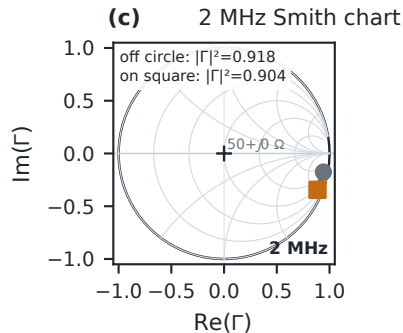
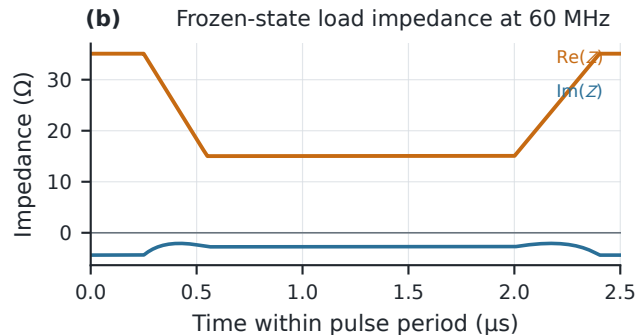
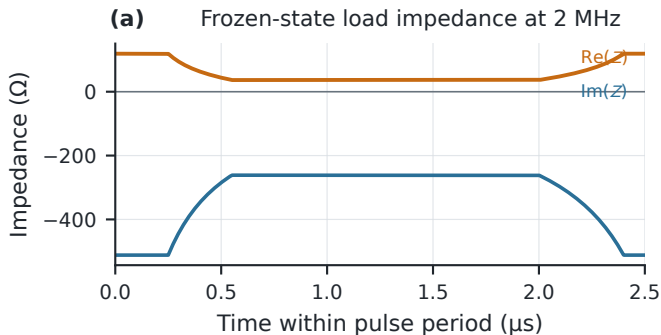
Voltage, current, charge and flux are compared directly over one complete 400 kHz pulse period.



Benchmark rule: all five RMS-normalized errors <1% and a complete ngspice transient. Wafer=0.019%; reflection=0.245%; all five pass. This establishes circuit-equation conformance, not chamber or plasma-chemistry validation.

Dynamic plasma load: impedance and 50 Ω Smith-chart interpretation

$Z_{load} = Z_{s,u} + R_p + j\omega L_p + Z_{s,w}$ and $\Gamma = (Z_{load} - 50)/(Z_{load} + 50)$; the external matcher is excluded.



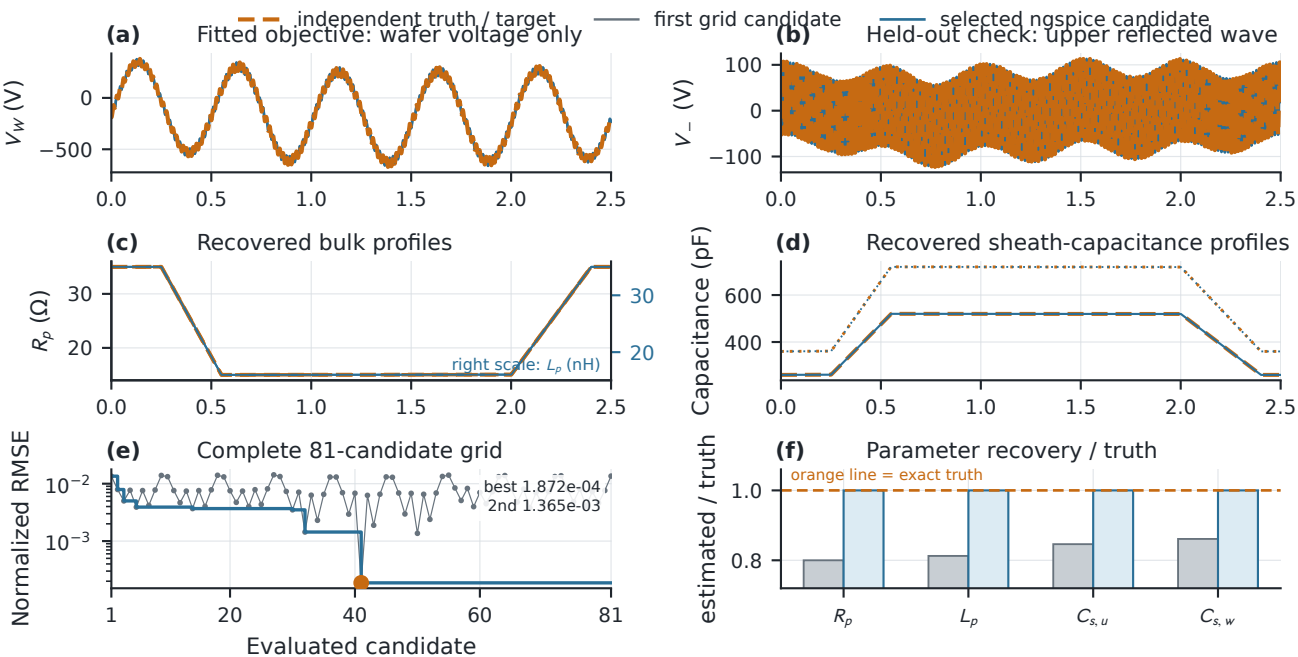
Smith center = 50 + j0 Ω (no reflection); outer circle = complete reflection.

Lower half = capacitive; gray circle is pulse-off and orange square is on-state.

These are frozen-state plasma-load values before matching, not measured source-plane S11 and not a kinetic plasma solution.

Inverse validity: one measured waveform recovers four bounded on-state values

Only wafer voltage contributes to loss; upper reflection is retained as a genuinely unused circuit-level validation output.



Selected: $R_p=15 \Omega$, $L_p=16 \text{ nH}$, $C_{su}=520 \text{ pF}$, $C_{sw}=720 \text{ pF}$. Objective nRMSE=1.872e-04; held-out nRMSE=2.452e-03; second-best / best loss=7.29.

Evidence basis, realism assessment, and model boundary

Published apparatus topology and scale checks are separated from synthetic benchmark inputs.

(a) Primary-source basis

[R1] H. J. Kim et al., JJAP 54, 01AE07 (2015)

300 mm DF-CCP; 20 mm gap; upper 13.56-60 MHz; lower 2 MHz.
doi:10.7567/JJAP.54.01AE07

[R2] T. Yamaguchi et al., J. Phys. D 45, 025203 (2012)

Negative DC above about 800 V on the upper 60 MHz electrode;
lower electrode was 13.56 MHz. doi:10.1088/0022-3727/45/2/025203

[R3] J. Kim et al., Electronics 14, 2022 (2025)

CCP load interpreted as sheath capacitance plus plasma L and R;
50 ohm matching-network reference plane. doi:10.3390/electronics14102022

[R4] H. Lee et al., Rev. Sci. Instrum. 86, 083505 (2015)

Sub-microsecond V/I impedance observation in pulsed dual-frequency CCP.
doi:10.1063/1.4928121

[R5] F. Schmidt et al., Plasma Sources Sci. Technol. 27, 105017 (2018)

Generator, matcher, discharge and stray elements must be coupled explicitly.
doi:10.1088/1361-6595/aae429

(c) Transparent order-of-magnitude checks

Sheath C -> thickness

0.87-2.41 mm

$d_s = \epsilon_0 A / C_s$, 300 mm plate

Plasma L -> density

$2.58\text{e}+14\text{-}5.65\text{e}+14\text{ m}^{-3}$

$L_p = m_e l / (n_e e^2 A)$, $l = 18\text{ mm}$

R/L -> collision scale

$9.38\text{e}+08\text{-}1.00\text{e}+09\text{ s}^{-1}$

$\nu_m \approx R_p / L_p$

(b) What the numerical values mean

Literature-grounded

300 mm geometry, 20 mm gap, 60/2 MHz arrangement, upper negative DC, R-L-C plasma interpretation

Scale-grounded

Hundreds-of-volts RF/DC excitation, millimeter-order sheath thickness, 50 ohm generator reference

Benchmark-specific

400 kHz DC-pulse envelope, ideal source amplitudes, matcher/feed/stray values, exact R/L/C trajectories

Identification-specific

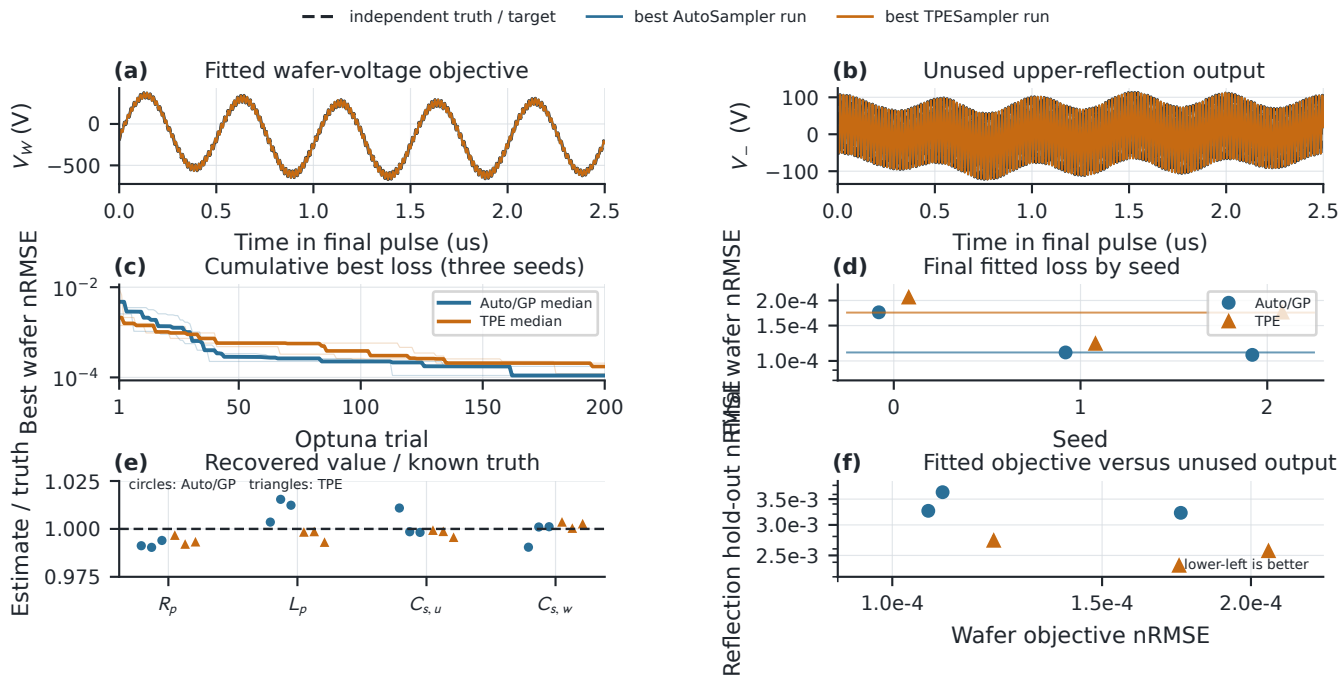
Three values for each of four on-state parameters; complete $3^4 = 81$ candidate enumeration

Not established

Gas chemistry, electron-energy distribution, ion-energy distribution, etch rate, uniformity, or a calibrated production recipe

OptunaHub AutoSampler (GP) versus standard TPE at 200 trials

Three matched seeds; the wafer waveform is fitted and upper reflection is retained as an unused output.



Fitted waveform: median nRMSE Auto/GP 1.102×10^{-4} , TPE 1.741×10^{-4} (ratio 0.633).

Unused reflection: median nRMSE Auto/GP 3.263×10^{-3} , TPE 2.564×10^{-3} . Execution: 1200 trials, 1150 new solves, 50 cache hits, 0 failures.

Assessment: Auto/GP improves the fitted objective, but repeated proposals and worse hold-out prevent a general superiority claim.